



Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models

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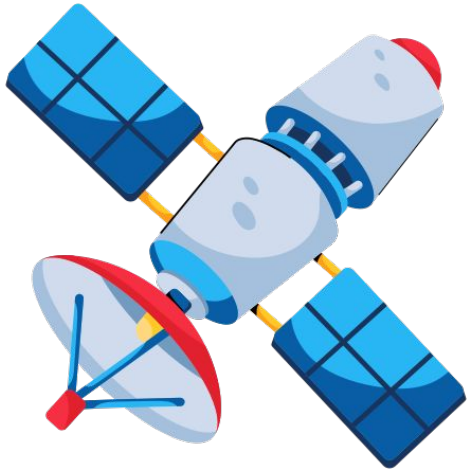
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GEAR Lab Website: <https://www.geoeearlab.com>

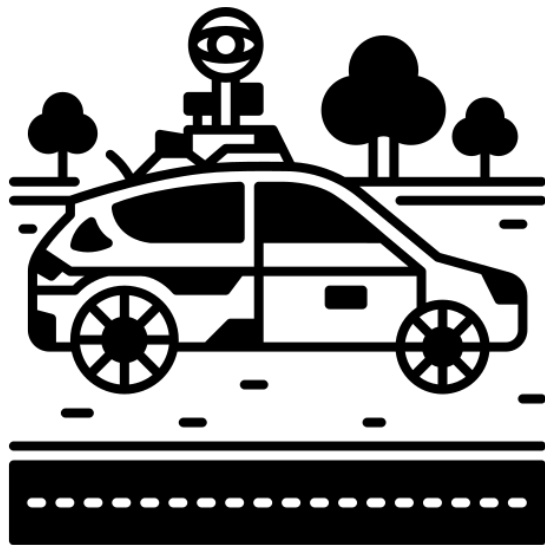


Background



Satellite Imagery--Rapid, large-scale observation immediately after disasters

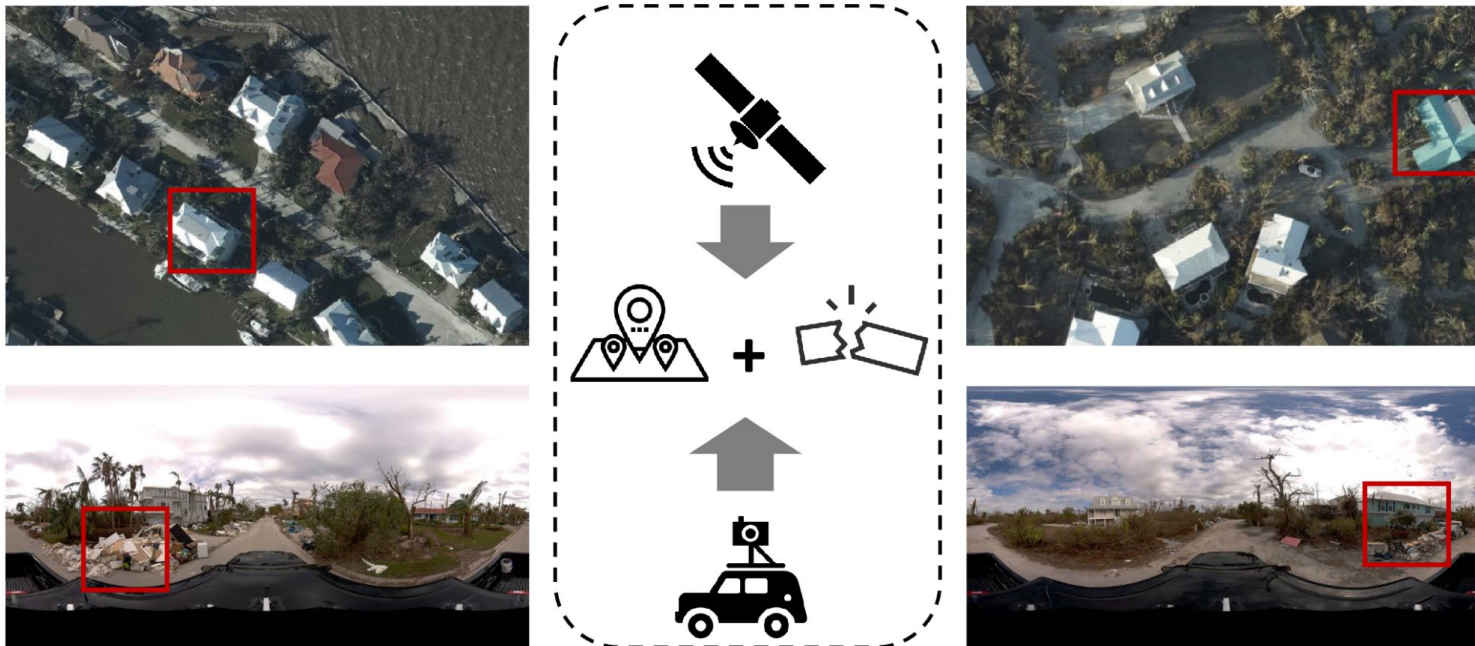
Background



Street-View Imagery-- Human-scale context, essential for structural damage assessment

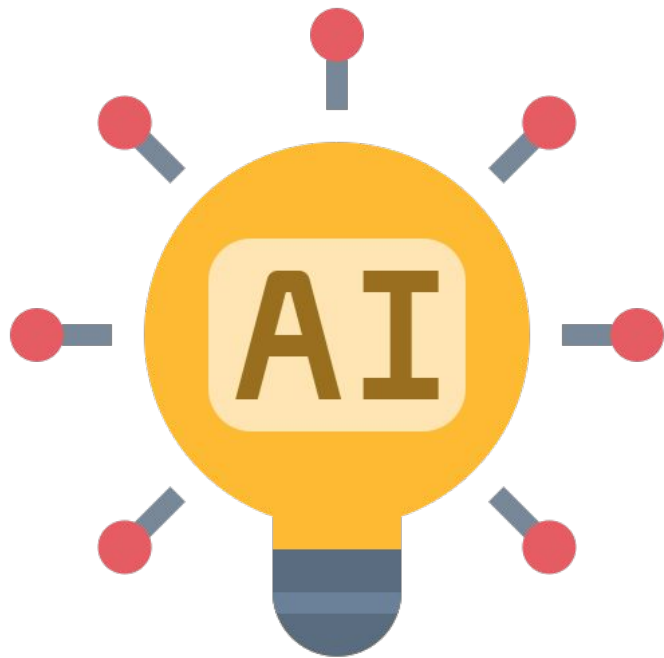
Background

The Data Gap: ground-level evidence is most valuable exactly when it is hardest to collect.



Li, H., Deuser, F., Yina, W., Luo, X., Walther, P., Mai, G., Huang, W., & Werner, M. (2024). Cross-View Geolocation and Disaster Mapping with Street-View and VHR Satellite Imagery: A Case Study of Hurricane IAN. *ArXiv*.
<https://arxiv.org/abs/2408.06761>

Background



Recent work uses street-view for hyperlocal damage classification & cross-view geolocalization —

but generating post-disaster street views from satellite imagery remains unexplored.

Generative AI



Research Questions

- RQ1: Can generative vision models synthesize geometrically consistent, semantically accurate post-disaster street views from satellite imagery?



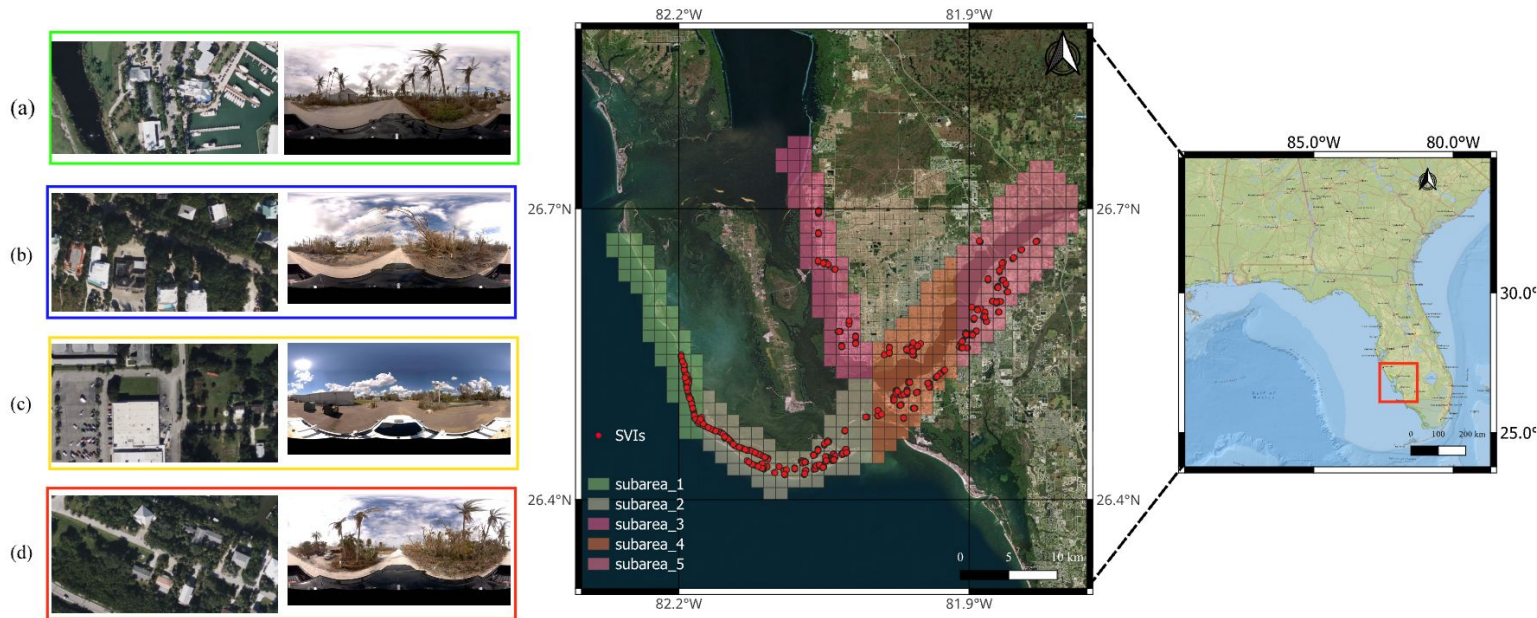
- RQ2: Does injecting explicit disaster semantics (VLM-guided prompts / severity-specialized MoE) improve damage fidelity?



- RQ3: How should we evaluate trustworthiness beyond pixel metrics?
→ Structure-Aware Evaluation Framework

Problem Definition & Dataset

Explore Cross-View Image Synthesis (CVIS) by learning a mapping function $G : I_{sat} \rightarrow I_{street}$ to directly synthesize high-quality, geometrically consistent, and semantically accurate post-disaster street-view imagery from post-disaster satellite observations.



Li, H., Deuser, F., Yina, W., Luo, X., Walther, P., Mai, G., Huang, W., & Werner, M. (2024). Cross-View Geolocalization and Disaster Mapping with Street-View and VHR Satellite Imagery: A Case Study of Hurricane IAN. *ArXiv*.
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Methodology

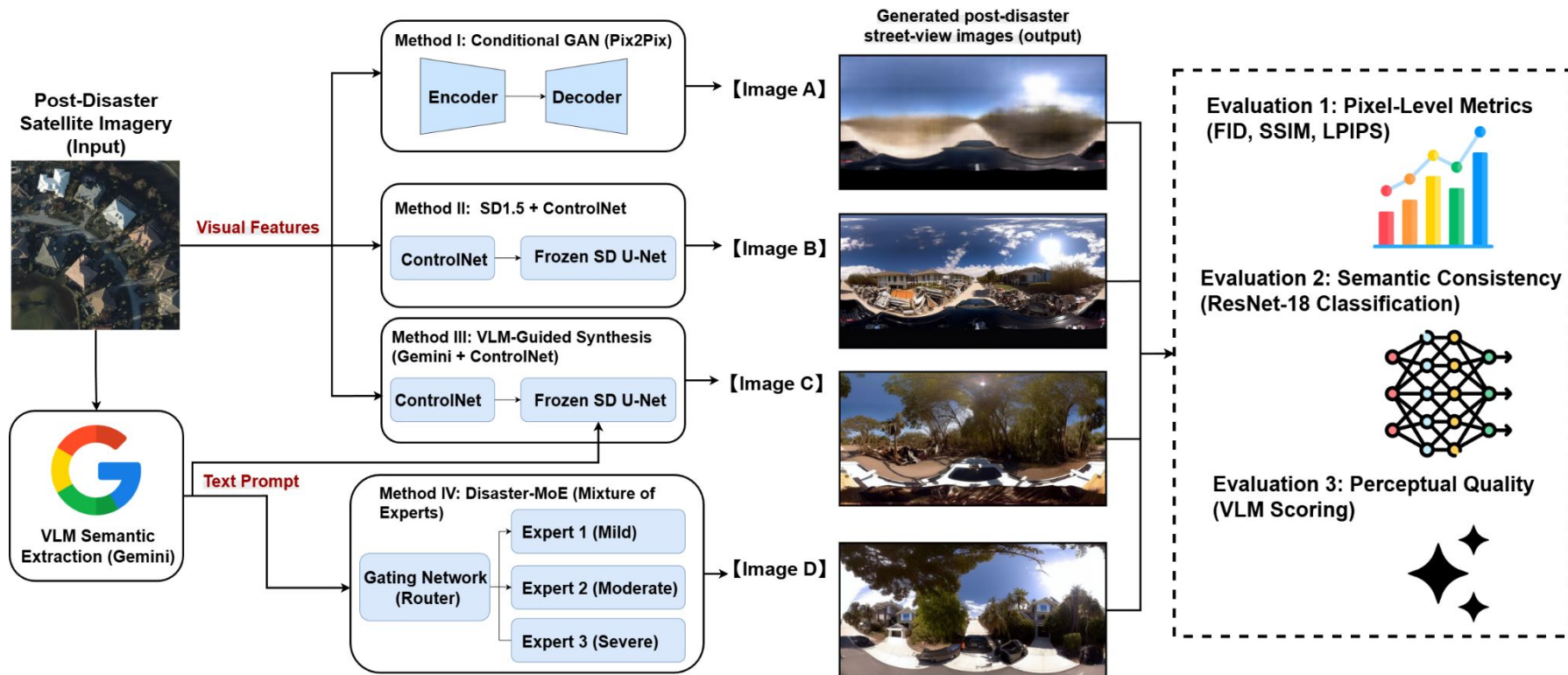
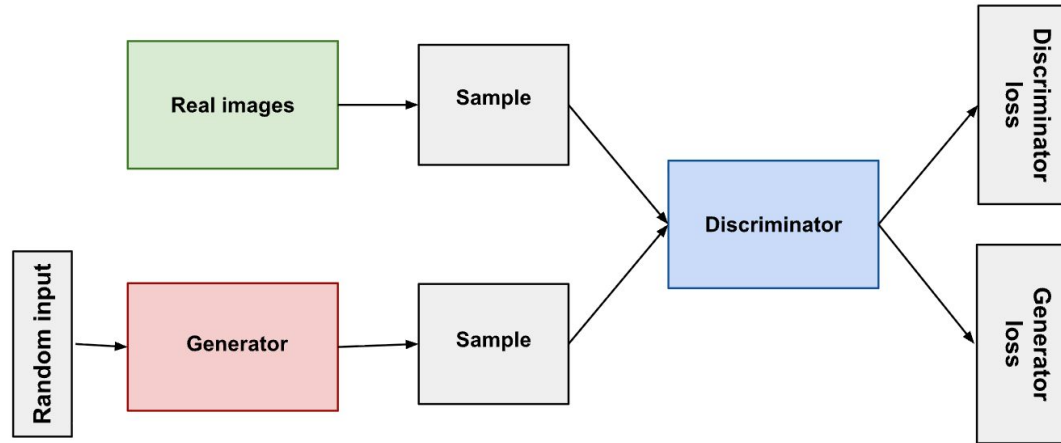


Figure. Overview of the proposed framework.

Methodology--Conditional GAN (Pix2Pix)

Research Philosophy: End-to-End Direct Mapping

Focuses on standard supervised image-to-image translation. It assumes a direct deterministic/adversarial mapping between satellite visual features and street-view perspective without intermediate high-level semantic reasoning.

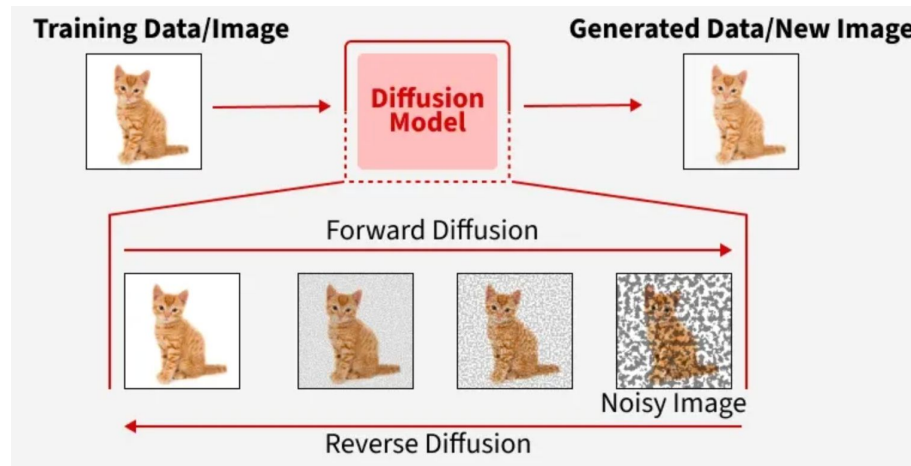


https://developers.google.com/machine-learning/gan/gan_structure

Methodology--SD1.5 + ControlNet

Research Philosophy: Pre-trained Diffusion Prior + Structural Control

This approach utilizes the rich prior generative capabilities of large-scale pre-trained diffusion models (SD 1.5) while using ControlNet to inject fine-grained spatial and structural visual conditions from satellite imagery.

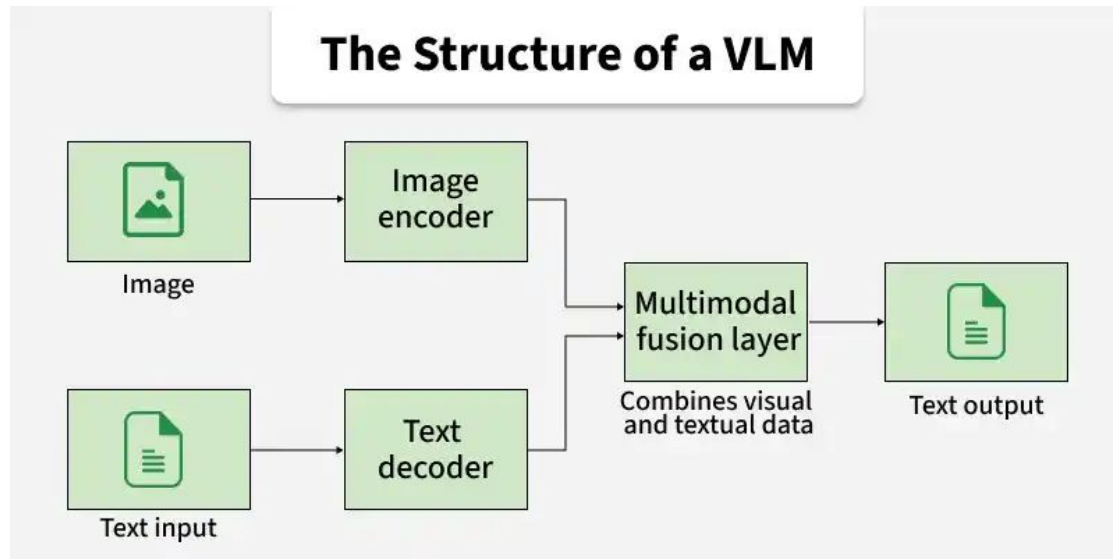


https://www.geeksforgeeks.org/artificial-intelligence/diffusion-models-in-machine-learning/#google_vignette

Methodology--VLM-Guided Synthesis (Gemini + ControlNet)

Research Philosophy: Multimodal Cross-Domain Semantic Conditioning

Bridges high-level visual understanding with generative diffusion. By extracting textual semantics via a Vision-Language Model (Gemini), it enhances the spatial visual features with rich textual context for dual-conditioned image synthesis.

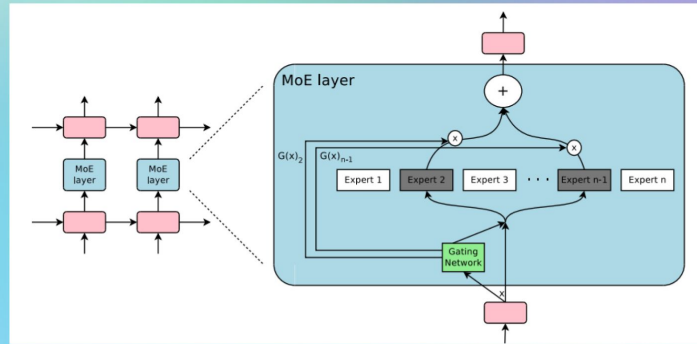


Methodology--Disaster-MoE (Mixture of Experts)

Research Philosophy: Dynamic Divide-and-Conquer & Specialized Routing

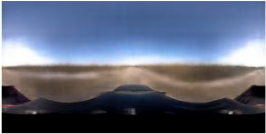
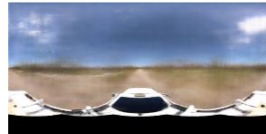
Replaces the "one-size-fits-all" model with domain specialization. A gating network dynamically routes tasks to specialized expert models (Mild, Moderate, Severe) based on VLM semantic analysis, achieving higher fidelity in nuanced disaster conditions.

Hugging Face explains Mixture of Experts (MoE)



<https://huggingface.co/blog/moe>

Results

	Satellite	Pix2Pix	SD1.5+ControlNet	SD1.5+ControlNet+Gemini	MoE	Ground Truth
Mild						
Moderate						
Severe						

Results

Table 1 Quantitative Comparison of Perceptual Image Quality Metrics

Method	SSIM(↑)	PSNR(↑)	LPIPS(↓)	FID(↓)
Pix2Pix	0.58609614	15.30729573	0.549302574	150.8341348
SD1.5_ControlNet	0.31436218	9.809302381	0.601944895	74.32713682
SD1.5_ControlNet_Gemini	0.290885676	9.726198367	0.603852578	82.19407033
MoE	0.221760725	8.453305609	0.6879829	134.522164
LoRA	0.1312	8.9619	0.6223	171.5578

- Pixel level: Pix2Pix wins SSIM/PSNR but has the worst FID (150.8) — low-frequency layouts without disaster texture

Results

Table 2 Classification Accuracy Score for Disaster Severity Consistency

Method	Accuracy	Recall	Precision	F1_Score	Mild_Acc	Mod_Acc	Sev_Acc
GroundTruth	<u>0.73</u>	<u>0.73</u>	<u>0.76</u>	<u>0.74</u>	<u>0.77</u>	<u>0.76</u>	<u>0.66</u>
Pix2Pix	0.34	0.34	0.28	0.17	1	0.01	0
SD1.5_Control Net	0.72	0.72	0.74	0.71	0.91	0.4	0.86
SD1.5_Control Net_Gemini	0.43	0.43	0.45	0.43	0.4	0.39	0.5
MoE	0.43	0.43	0.45	0.44	0.41	0.47	0.42
LoRA	0.37	0.37	0.39	0.34	0.27	0.12	0.73

Semantic consistency: standard ControlNet achieves the best F1 (0.71, near ground-truth bound 0.74; Severe acc 0.86) — rigid structural constraints preserve damage features

Results

Table 3 Semantic & Perceptual Evaluation by LLM

Method	Struct	Damage	Realism
Pix2Pix	1.26	1.08	1.00
SD1.5_ControlNet	1.43	1.68	2.11
SD1.5_ControlNet_Ge mini	1.88	2.04	2.08
MoE	1.61	1.79	2.11

Perceptual (VLM-as-a-Judge): VLM-guided synthesis scores best on Structural (1.88) & Damage Accuracy (2.04); MoE ties best Realism (2.11) — explicit semantic guidance bridges hallucination and reality

Takeaways

- Single-model approaches can't yet balance visual plausibility with strict structural alignment
- High perceptual realism $\neq$ high semantic accuracy — evaluation must be structure-aware
- Trustworthiness, not fidelity alone, should be the target for disaster generation



Yang, Y., Zou, L., & Jepson, W. (2026). Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models. *ArXiv*.
<https://arxiv.org/abs/2603.20697>

Future Directions

From View Translation to 3D Scene Generation

- Satellite (top-down) and street-view (ground-level) are complementary projections of the same post-disaster 3D scene — treating them as a 2D-to-2D translation problem forces the model to hallucinate geometry
- Next step: fuse both views to reconstruct / generate the 3D disaster scene (e.g., 3D Gaussian Splatting, NeRF, geometry-grounded diffusion), from which any viewpoint can be rendered with guaranteed cross-view consistency
- Industry momentum: Google Earth's Nano Banana grounds generation in satellite + 3D terrain data — validating geospatially-grounded generation, while its 1-day rollback underscores that trustworthy generation remains the open problem our evaluation framework targets

The Nano Banana AI Image Generator Has Arrived in Google Earth

NEWS JUL 31, 2026 MATT GROWCOOT

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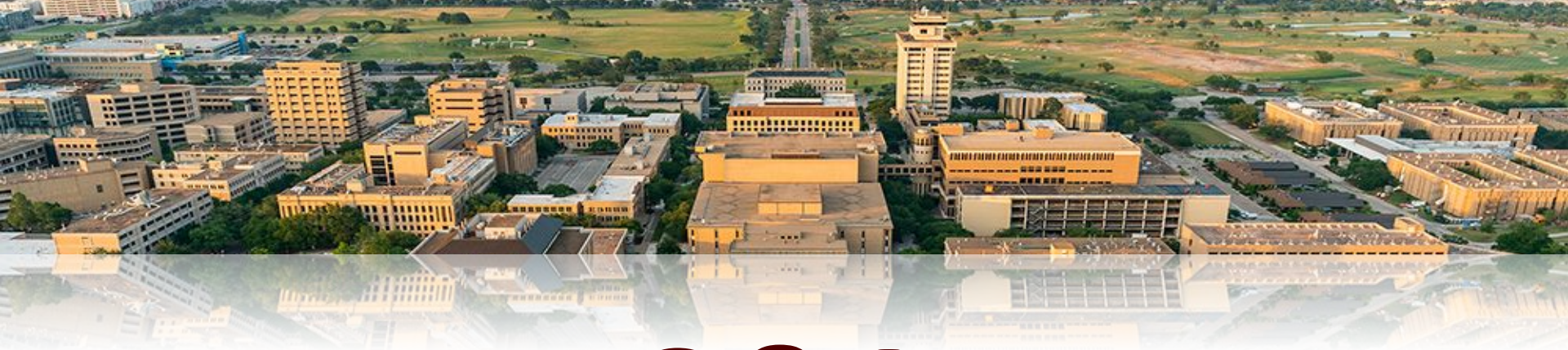


The original satellite image, left, was transformed into an imagining of what Pompeii looked like before Mount Vesuvius exploded.

Future Directions

Sources:

- Transform any place with Nano Banana in Google Earth — Google
(<https://blog.google/products-and-platforms/products/earth/nano-banana-google-earth-image-generation/>)
- Nano Banana 2 removed from Google Earth — Digital Digging
(<https://www.digitaldigging.org/p/nano-banana-2-removed-from-google/>)
- The Nano Banana AI Image Generator Has Arrived in Google Earth — PetaPixel
(<https://petapixel.com/2026/07/31/the-nano-banana-ai-image-generator-has-arrived-in-google-earth/>)
- Google's AI satellite image tool lasted exactly one day — Crypto Briefing
(<https://cryptobriefing.com/google-ai-satellite-image-retracted/>)



Q&A

Welcome any comments/questions
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